" If You Want To Manage It, Measure It "

IOTAFLOW
SYSTEMS PVT. LTD

OVAL GEAR FLOW METERS

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Principle Of Oval Gear Flow meter Operation

The oval gear flow meter is a kind of Positive Displacement Flow Meter. As the fluid being measured passes thru the meter, it rotates 2 oval gears in measuring cavity to displace a precise volume of fluid. A sensor detects the gear rotation to determine displaced volume and flow rate.

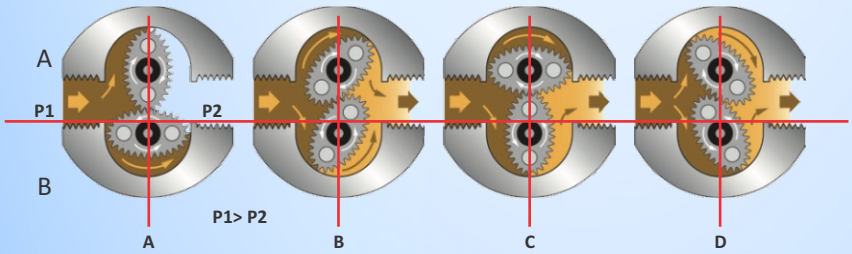


Figure 1: The Principle of Oval Gear Operation

Fluid pressure rotates the oval gear, Figure 1. In position (a), the fluid exerts a clockwise driving force on gear A. there is no driving force on gear B. It is perpendicular to the flow so the fluid forces are balanced around the shaft. As the gear rotate to position (b), the fluid being to exert a force on gear B. At position (c), all the driving force is on the gear B. This alternating driving force provides a smooth rotation of almost constant torque.

The meter design minimizes the slippage between the gears and the measuring cavity wall. As a result, the oval meter is less affected than other designs by the liquid's viscosity and lubricity.

Application

Oval gear flow meter is mainly used for high value liquids such as oil, chemical etc. Main applications include unloading, transfer and consumption monitoring.

Oval Gear Flow meter Features

Only Two Moving Parts,
Higher Accuracy, 0.5% or 0.2%.
Standard Viscosity up to 5,000 cP (centipoise = MPa.s).
Size of wide flow ranges 0.04-340 M3/h (18-37,396GPM).
Wide range of working environment –20 ~ +280 °C

Body available Cast Iron, Cast Steel, and Stainless Steel.
With LC Display, Transmitter, MCU Flow process controller option.

Model No.		OGM - RA		OGM - RB			OGM - RC	
Body material		Cast Iron		Cast Steel			Stainless Steel	
Pressure (MPa)		1.0	1.6	2.5	4.0	6.4	1.0	1.6
Liquid Temperature		-20°C ~ +100°C						
Flange		#150-300 PN ANSI						
Liquid Viscosity (MPa-s)		0.3~2 MPa.s		2~8 MPa.s			8~200 MPa.s	
Flowmeter (m^3/h)	DN 10mm	0.2~0.4	0.08 ~0.4	0.1~0.4	0.08~0.4		0.08	0.04~0.4
	DN 15mm	0.75~1.5	0.3~1.5	0.3~1.5	0.3~1.5		0.3~1.5	0.15~1.5
	DN 20mm	1~3	0.4~3	0.6~3	0.4~3		0.5~3	0.3~3
	DN 25mm	2~6	0.8~6	1.2~6	0.8~6		1~6	0.6~6
	DN 40mm	5~15	2~15	3~15	2~15		2.5~15	1.5~15
	DN 50mm	8~24	3~24	4.8~24	3~24		4~24	2.4~24
	DN 65mm	15~40	5~40	8~40	5~40		6~40	4~40
	DN 80mm	20~60	6~60	12~60	6~60		10~60	6~60
	DN 100mm	34~100	10~100	20~100	10~100		16~100	10~100
	DN 150mm	64~190	19~190	38~190	19~190		32~190	19~190
	DN 200mm	114~340	34~340	68~340	34~340		56~340	34~340
Accuracy		0.2%	0.5%	0.2%	0.5%		0.2%	0.5%

OGM- T Series Higher Temperature Oval Gear Flow Meter

Model No.		OGM - TA		OGM - TB			OGM - TC	
Body material		Cast Iron		Cast Steel			Stainless Steel	
Pressure (MPa)		1.0	1.6	2.5	4.0	6.4	1.0	1.6
Liquid Viscosity		2 - 200 MPa_s						
Liquid Temperature		+100°C ~ +280°C						
Flange		#150			#300			
Accuracy		0.5%	0.2%	0.5%	0.2%		0.5%	0.2%
Flow Range (m^3 /h)	DN 10mm	0.04-0.4	0.1-0.4	0.04-0.4	0.1-0.4		0.04-0.4	0.1-0.4
	DN 15mm	0.24-1.35	0.35-1.35	0.24-1.35	0.35-1.35		0.24-1.35	0.35-1.35
	DN 20mm	0.54-2.7	0.72-2.7	0.54-2.7	0.72-2.7		0.54-2.7	0.72-2.7
	DN 25mm	1.2-5.4	1.4-5.4	1.2-5.4	1.4-5.4		1.2-5.4	1.4-5.4
	DN 40mm	2.7-3.5	3.6-13.5	2.7-3.5	3.6-13.5		2.7-3.5	3.6-13.5
	DN 50mm	4.4-21.6	5.75-21.6	4.4-21.6	5.75-21.6		4.4-21.6	5.75-21.6
	DN 65mm	8.4-36	9.6-36	8.4-36	9.6-36		8.4-36	9.6-36
	DN 80mm	10.8-54	14.4-54	10.8-54	14.4-54		10.8-54	14.4-54
	DN 100mm	18-90	24-90	18-90	24-90		18-90	24-90
	DN 150mm	38-170	45.6-170	38-170	45.6-170		38-170	45.6-170
	DN 200mm	34-340	68-340	34-340	68-340		34-340	68-340

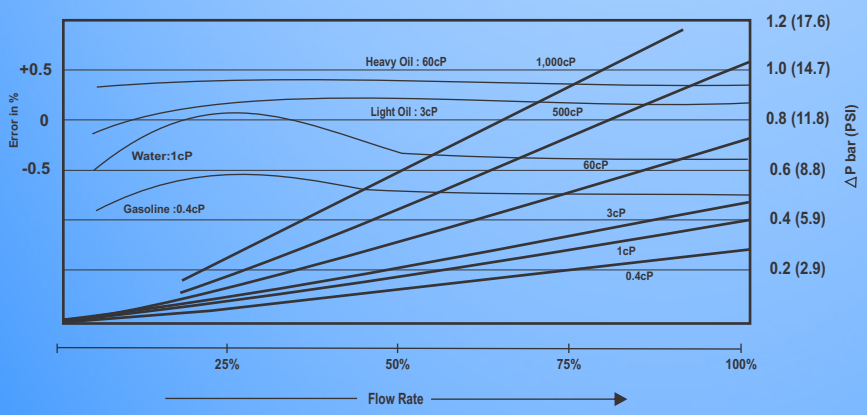
OGM- N Series Higher Viscosity Oval Gear Flow Meter

Model No.		OGM - NA			OGM - NB			OGM - NC		
Body material		Cast Iron			Cast Steel			Stainless Steel		
Pressure (MPa)		1.6			2.5	4.0	6.4	1.6		
Liquid Temperature		-10°C ~ +100°C								
Flange		# 150			# 300					
Liquid Viscosity (MPa-s)		200-1,000		1,000-5,000	200~1,000		1,000~5,000	200~1,000		1,000~5,000
Flowmeter (m^3 /h)	DN 10mm	0.04-0.2	0.04-0.2	0.04-0.2	0.04-0.2	0.04-0.2	0.04-0.2	0.04-0.2	0.04-0.2	0.04-0.2
	DN 15mm	0.2-1.0	0.1-1.05	0.075-0.75	0.2-1.0	0.1-1.05	0.075-0.75	0.2-1.0	0.1-1.05	0.075-0.75
	DN 20mm	0.42-2.1	0.21-2.1	0.15-1.5	0.42-2.1	0.21-2.1	0.15-1.5	0.42-2.1	0.21-2.1	0.15-1.5
	DN 25mm	0.84-4.2	0.42-4.2	0.3-3	0.84-4.2	0.42-4.2	0.3-3	0.84-4.2	0.42-4.2	0.3-3
	DN 40mm	2.1-10.5	1.05-10.5	0.75-7.5	2.1-10.5	1.05-10.5	0.75-7.5	2.1-10.5	1.05-10.5	0.75-7.5
	DN 50mm	3.36-16.8	1.68-16.8	1.2-12	3.36-16.8	1.68-16.8	1.2-12	3.36-16.8	1.68-16.8	1.2-12
	DN 65mm	5.6-28	2.8-28	2-20	5.6-28	2.8-28	2-20	5.6-28	2.8-28	2-20
	DN 80mm	8.4-42	4.2-42	3-30	8.4-42	4.2-42	3-30	8.4-42	4.2-42	3-30
	DN 100mm	14-70	6.70	5-5-	14-70	6.70	5-5-	14-70	6.70	5-5-
	DN 150mm	26.6-13.3	13.3-133	9.5-95	26.6-13.3	13.3-133	9.5-95	26.6-13.3	13.3-133	9.5-95
	DN 200mm	47.6-23.8	23.8-238	17-170	47.6-23.8	23.8-238	17-170	47.6-23.8	23.8-238	17-170
Accuracy		0.2%	0.5	0.5	0.2%	0.5	0.5	0.2%	0.5	0.5

We invite your inquires for flow meter for Oil. Water, Gas, Air, Chemicals & Effluents:

FLOW METERS	
TYPE	APPLICATION
• Variable Area	• Water
• Turbine	• Air
• Positive Displacement	• Oil
• Electromagnetic	• Gas
• Ultrasonic	• Chemicals
• RPD	• Effluent
• Diaphragm	

Accuracy & Pressure Drop Curves



2021	IOTAFLOW Make Oval Gear Flow Meters	Data Sheet Ver 1.1

	Oval Gear Meter-OGM	XX	X	X	X	X	X	X	X	X	X	X	X	X
OG Size	OG DN10	10												
	OG DN15	15												
	OG DN20	20												
	OG DN25	25												
	OG DN40	40												
	OG DN50	50												
	OG DN65	65												
	OG DN80	80												
	OG DN100	100												
	OG DN150	150												
	OG DN200	200												
Display Type	OG Digital Flow Rate & Totalizer		D											
	OG Digital Flow Rate		F											
	OG Mechanical		M											
	OG Mechanical Resettable		M1											
Body	OGB Cast Iron			D										
	OGB SS 316			F										
	OGB SS 304			S										
	OGB Cast Steel			C										
Gears	OGG Aluminium				A									
	OGG Cast Iron				D									
	OGG SS 304				S									
	OGG SS 316				F									
	OGG Cast Steel				C									
OG Service	OG Liquid					L								
	OG Oil					O								
OG Pressure Rating	OG 64 Bar						M							
	OG 40 bar						S							
	OG 25 bar						F							
	OG 16 Bar						P							
	OG 10 Bar						N							
	OG 6 Bar						B							
Seals	OG Neoprene							N						
	OG FFKM							K						
	OG PTFE							P						
	OG Viton							V						
	OG FPM							F						
OG End Connection	OG Flanged PN64								PN64					
	OG Flanged PN40								PN40					
	OG Flanged PN25								PN25					
	OG Flanged PN16								PN16					
	OG Flanged Class #300								N					
	OG Flanged Class #150								M					
Accuracy	OG 0.2%									C				
	OG 0.5%									B				
	OG 1%									A				
OG Output	OG 4-20 MA										A			
	OG Pulse										D			
	OG No Output										O			
Temp Suitability	OG 100 Deg C											100		
	OG 150 Deg C											150		
	OG 280 Deg C											280		
OG Display Enclosure	OG IP 63												A	
	OG IP 65												B	
	OG IP 67												C	
	OG IP 68												D	
	OG Nema-4x												N	
	OG Ex-Proof												E	
OG Power Supply	OG Skip													O
	OG Battery													B
	OG 24 VDC													24
	OG 220 VAC 50 Hz													220
OG Series	OG High Viscosity														N
	OG High Temperature														T
	OG Normal														R

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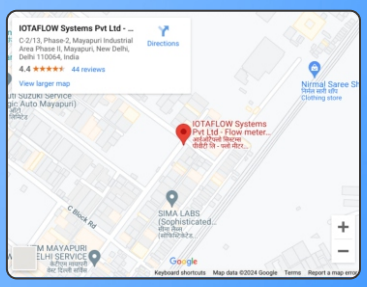

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